



**DEPARTMENT OF ARTIFICIAL INTELLIGENCE  
AND DATA SCIENCE**

**101008/IT722T - Seminar Diary  
2022-26 Batch  
JULY-NOVEMBER 2025**

**Citation & Retraction Patterns in Paper Mill  
Publications**

**Guided by  
Dr. Varghese S. Chooralil**

**Submitted by  
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RET22AD041**

**Rajagiri School of Engineering & Technology (Autonomous)**

Rajagiri Valley, Kakkanad, Kochi, 682039

May 2025

# **RAJAGIRI SCHOOL OF ENGINEERING & TECHNOLOGY (AUTONOMOUS)**

## **Vision**

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

## **Mission**

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

## **DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE**

### **Vision**

To become a centre of excellence in Artificial Intelligence and Data Science and moulding professionals by imparting high-quality knowledge for the sustainable betterment of mankind.

### **Mission**

To empower individuals with expertise in constantly evolving AI and Data Science tools and technologies for impactful innovations, research and societal advancements through ethical practices and industrial collaborations.

### **Programme Outcomes (PO)**

Artificial Intelligence and Data Science Students will be able to:

**PO1: Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

**PO2: Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

**PO3: Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

**PO4: Conduct investigations of complex problems:** Use research-based knowledge including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

**PO5: Modern Tool Usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

**PO6: The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

**PO7: Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

**PO8: Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

**PO9: Individual and Team work:** Function effectively as an individual, and as a member or leader in teams, and in multidisciplinary settings.

**PO10: Communication:** Communicate effectively with the engineering community and with society at large. Be able to comprehend and write effective reports documentation. Make effective presentations, and give and receive clear instructions.

**PO11: Project management and finance:** Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work, as a member and leader in a team. Manage projects in multidisciplinary environments.

**PO12: Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.

### **Programme Specific Outcomes (PSO)**

Artificial Intelligence & Data Science Program Students will be able to:

**PSO1 :** Apply the fundamentals of science, engineering and mathematics to understand, analyze and develop solutions in the areas related to artificial intelligence and data science for optimal design of intelligent systems.

**PSO2 :** Design and Implement appropriate techniques and analytic tools for the integration of intelligent systems, with a view to engaging in lifelong learning for the betterment of society.

**PSO3 :** Practice professional ethics in applying scientific method to model and support multidisciplinary facets of engineering and its societal implications.

### **Course Outcomes (CO)**

After completion of the course the student will be able to:

**CO1:** Analyse a current topic of professional interest and present it before an audience

**CO2:** Identify an engineering problem, analyse it and propose a work plan to solve it

**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE, RSET**  
**B.Tech. ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**

**100008/IT722T-SEMINAR**  
**S7 - AI&DS (2022-2026 Admission)**

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| <b>SEMINAR INFORMATION SUMMARY SHEET</b> |
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|----|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | Title of the Seminar    | <b>Citation &amp; Retraction Patterns in Paper Mill Publications</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 2. | Name of the Student     | <b>Liz Maria Benny</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 3. | Name of the Guide       | <b>Dr. Varghese S. Chooralil</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 4. | Abstract of the Seminar | Paper mills mass-produce fraudulent scientific manuscripts, often infiltrating reputable journals and polluting the integrity of the scholarly record. This seminar analyzes retraction and citation patterns in 416 retracted paper-mill publications, encompassing over 14,000 references and 8,400 citations. The study reveals the existence of coordinated “citation rings”, delayed detection (median lag 3.4 years), and widespread contamination, including in Q2 journals. These findings—together with visible anomalies in image and style—motivate AI-powered integrity tools like VERITAS, which combine citation analytics, image forensics, and stylometry to defend science from networked fraud. |

**Signature of the Student**

**Signature of the Guide**

**Date:**

**RAJAGIRI SCHOOL OF ENGINEERING & TECHNOLOGY**  
**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**

Batch 2022-2026, JULY - NOVEMBER 2025

**SEMINAR - PROGRESS EVALUATION- WEEK 1**

|                      |                                                           |
|----------------------|-----------------------------------------------------------|
| <b>Seminar Title</b> | Citation & Retraction Patterns in Paper Mill Publications |
| <b>Date</b>          | 02 July 2025                                              |
| <b>Remarks</b>       |                                                           |

Name & Signature of Guide:

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**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**

Batch 2022-2026, JULY - NOVEMBER 2025

**SEMINAR - PROGRESS EVALUATION- WEEK 2**

|                      |                                                           |
|----------------------|-----------------------------------------------------------|
| <b>Seminar Title</b> | Citation & Retraction Patterns in Paper Mill Publications |
| <b>Date</b>          | 09 July 2025                                              |
| <b>Remarks</b>       |                                                           |

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**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**

Batch 2022-2026, JULY - NOVEMBER 2025

**SEMINAR - PROGRESS EVALUATION- WEEK 3**

|                      |                                                           |
|----------------------|-----------------------------------------------------------|
| <b>Seminar Title</b> | Citation & Retraction Patterns in Paper Mill Publications |
| <b>Date</b>          | 16 July 2025                                              |
| <b>Remarks</b>       |                                                           |

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Batch 2022-2026, JULY - NOVEMBER 2025

**SEMINAR - PROGRESS EVALUATION- WEEK 4**

|                      |                                                           |
|----------------------|-----------------------------------------------------------|
| <b>Seminar Title</b> | Citation & Retraction Patterns in Paper Mill Publications |
| <b>Date</b>          | 23 July 2025                                              |
| <b>Remarks</b>       |                                                           |

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Batch 2022-2026, JULY - NOVEMBER 2025

**SEMINAR - PROGRESS EVALUATION- WEEK 5**

|                      |                                                           |
|----------------------|-----------------------------------------------------------|
| <b>Seminar Title</b> | Citation & Retraction Patterns in Paper Mill Publications |
| <b>Date</b>          | 06 August 2025                                            |
| <b>Remarks</b>       |                                                           |

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Batch 2022-2026, JULY - NOVEMBER 2025

**SEMINAR - PROGRESS EVALUATION- WEEK 6**

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|----------------------|-----------------------------------------------------------|
| <b>Seminar Title</b> | Citation & Retraction Patterns in Paper Mill Publications |
| <b>Date</b>          | 13 August 2025                                            |
| <b>Remarks</b>       |                                                           |

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Batch 2022-2026, JULY - NOVEMBER 2025

**SEMINAR - PROGRESS EVALUATION- WEEK 7**

|                      |                                                           |
|----------------------|-----------------------------------------------------------|
| <b>Seminar Title</b> | Citation & Retraction Patterns in Paper Mill Publications |
| <b>Date</b>          | 20 August 2025                                            |
| <b>Remarks</b>       |                                                           |

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Batch 2022-2026, JULY - NOVEMBER 2025

**SEMINAR - PROGRESS EVALUATION- WEEK 8**

|                      |                                                           |
|----------------------|-----------------------------------------------------------|
| <b>Seminar Title</b> | Citation & Retraction Patterns in Paper Mill Publications |
| <b>Date</b>          | 27 August 2025                                            |
| <b>Remarks</b>       |                                                           |

Name & Signature of Guide: